

MCY 611 Cloud Computing Lab#3: AWS Elastic Block Store (EBS)

Part 1. Create a new EBS volume

1. Create an instance via the following (please note, you created a security group, keypair, and subnet before in lab 1, go back to that lab and retrieve those):

```
aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 1 --instance-type t3.micro --key-name  
your_user_name-key --security-group-ids your_security_group_id --subnet-id your_subnet_id
```

```
~ $ aws ec2 run-instances --image-id ami-09e67e426f25ce0d7 --count 1 --instance-type t3.micro --key-name carrd-key --security-group-ids sg-0dc79f8b11b3a6809 --subnet-id subnet-0aa6a451bcda4099f  
{  
  "ReservationId": "r-00fcacea4403bd9cf",  
  "OwnerId": "881490118823",  
}
```

What is your EC2 instance's AvailabilityZone? The AvailabilityZone should be something like us-east-xx, for example us-east-1a.

us-east-1a

2. Create a new EBS volume via the following command:

```
aws ec2 create-volume --size 1 --region us-east-1 --availability-zone your_EC2_instance_AvailabilityZone
```

What does this command mean? What is your command and its output?

```
~ $ aws ec2 create-volume --size 1 --region us-east-1 --availability-zone us-east-1a  
{  
  "Iops": 100,  
  "Tags": [],  
  "VolumeType": "gp2",  
  "MultiAttachEnabled": false,  
  "VolumeId": "vol-049e00e0e2d5824ce",  
  "Size": 1,  
  "SnapshotId": "",  
  "AvailabilityZone": "us-east-1a",  
  "State": "creating",  
  "CreateTime": "2025-01-31T19:48:54+00:00",  
  "Encrypted": false  
}
```

This command creates an EBS volume. The size of the volume created being 1 gigabyte and in the us-east-1 region.

3. Describes your EBS volume via the following command:

```
aws ec2 describe-volume-status --volume-ids volume_id
```

You can get the *volume_id* of the EBS volume from the output of step 2. **What is the output?**

```

~ $ aws ec2 describe-volume-status --volume-ids vol-049e00e0e2d5824ce
{
  "VolumeStatuses": [
    {
      "Actions": [],
      "AvailabilityZone": "us-east-1a",
      "Events": [],
      "VolumeId": "vol-049e00e0e2d5824ce",
      "VolumeStatus": {
        "Details": [
          {
            "Name": "io-enabled",
            "Status": "passed"
          },
          {
            "Name": "io-performance",
            "Status": "not-applicable"
          }
        ],
        "Status": "ok"
      }
    }
  ]
}
~ $

```

4. Attach the EBS volume to the instance and exposes it as /dev/sdf via the following:

```
aws ec2 attach-volume --volume-id volume_id --instance-id instance_id --device /dev/sdf
```

What does this command mean? What is the output?

```

~ $ aws ec2 attach-volume --volume-id vol-049e00e0e2d5824ce --instance-id i-0bfe68e2519ab089b --device /dev/sdf
{
  "VolumeId": "vol-049e00e0e2d5824ce",
  "InstanceId": "i-0bfe68e2519ab089b",
  "Device": "/dev/sdf",
  "State": "attaching",
  "AttachTime": "2025-01-31T19:53:27.426000+00:00"
}
~ $

```

5. Log into the EC2 instance via ssh. Switch to root. You have learned how to log into an EC2 instance in lab1.

What are your commands and their outputs?

```

~ $ ssh -i devenv-key.pem ubuntu@54.174.77.210
The authenticity of host '54.174.77.210 (54.174.77.210)' can't be established.
ED25519 key fingerprint is SHA256:9Bt1vdTTJTt5YN7aY9hpeS6hB697nzexKKqxHYcfuyc.
This key is not known by any other names
Are you sure you want to connect to 54.174.77.210 (yes/no/[fingerprint])? yes

```

```

ubuntu@ip-172-31-42-75:~$ sudo -s
root@ip-172-31-42-75:/home/ubuntu#

```

Once you've successfully logged into your EC2 instance, your CloudShell terminal becomes an EC2 instance terminal. You will see the command prompt change from [cloudshell-user@...] to root@....

Create a file system on the EBS volume and mount the volume via the following commands. Note that depending on the block device driver of the kernel, the device might be attached with a different name than what you specify. For example, if you specify a device name of /dev/sdf, your device might be renamed /dev/nvme1n1 by the kernel. Run “lsblk” on the terminal to view your available disk devices and their mount points (if applicable) to help you determine the correct device name to use. Assume that the device is renamed /dev/nvme1n1 by the kernel.

```

root@ip-172-31-42-75:/home/ubuntu# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0        7:0      0  33.3M  1 loop /snap/amazon-ssm-agent/3552
loop1        7:1      0  55.5M  1 loop /snap/core18/1997
loop2        7:2      0  70.4M  1 loop /snap/lxd/19647
loop3        7:3      0  32.3M  1 loop /snap/snapd/11588
nvme0n1      259:0    0    8G   0 disk 
└─nvme0n1p1 259:1    0    8G   0 part /
root@ip-172-31-42-75:/home/ubuntu# mkfs -F /dev/nvme0n1
mkfs 1.45.5 (07-Jan-2020)
Found a dos partition table in /dev/nvme0n1
/dev/nvme0n1 is apparently in use by the system; will not make a filesystem here!

```

The newly created volume doesn't have a file system. So you need to create a file system on the volume using the mkfs command before you can use it.

```
mkfs -F /dev/nvme1n1
```

What is the output?

```

root@ip-172-31-42-75:/home/ubuntu# mount /dev/nvme0n1 /data
mount: /data: /dev/nvme0n1 already mounted or mount point busy.
root@ip-172-31-42-75:/home/ubuntu# cd /data/
bash: cd /data/: No such file or directory
root@ip-172-31-42-75:/home/ubuntu# df

```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
/dev/root	8065444	1336428	6712632	17%	/
devtmpfs	482284	0	482284	0%	/dev
tmpfs	488072	0	488072	0%	/dev/shm
tmpfs	97616	776	96840	1%	/run
tmpfs	5120	0	5120	0%	/run/lock
tmpfs	488072	0	488072	0%	/sys/fs/cgroup
/dev/loop0	34176	34176	0	100%	/snap/amazon-ssm-agent/3552
/dev/loop2	72192	72192	0	100%	/snap/lxd/19647
/dev/loop1	56832	56832	0	100%	/snap/core18/1997
/dev/loop3	33152	33152	0	100%	/snap/snapd/11588
tmpfs	97612	0	97612	0%	/run/user/1000

Once you've created the file system, you can mount the volume using the mount command.

```
mkdir /data
```

```
mount /dev/nvme1n1 /data
```

```
cd /data/
```

```
df
```

What is the output?

```

root@ip-172-31-42-75:/home/ubuntu# df

```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
/dev/root	8065444	1336428	6712632	17%	/
devtmpfs	482284	0	482284	0%	/dev
tmpfs	488072	0	488072	0%	/dev/shm
tmpfs	97616	776	96840	1%	/run
tmpfs	5120	0	5120	0%	/run/lock
tmpfs	488072	0	488072	0%	/sys/fs/cgroup
/dev/loop0	34176	34176	0	100%	/snap/amazon-ssm-agent/3552
/dev/loop2	72192	72192	0	100%	/snap/lxd/19647
/dev/loop1	56832	56832	0	100%	/snap/core18/1997
/dev/loop3	33152	33152	0	100%	/snap/snapd/11588
tmpfs	97612	0	97612	0%	/run/user/1000

Note: If the EBS volume you're mounting already has a file system, then you won't need to run mkfs on it. In that case, you can simply mount the volume using the mount command.

Part 2. EBS Snapshots

6. Create a file named mcy611.txt in the /data directory that was created in step 5 and type anything you want in the file. Let's create a backup of your EBS volume. The first step is to make sure that any in-memory data has been written to the volume (disk). Run the sync command in the terminal of the EC2 instance. The following is how I ran the sync command.

```
root@ip-10-45-185-154:/data# sync
```

7. **Open a new CloudShell terminal.** Create a snapshot (backup) of your EBS volume via the following:

```
aws ec2 create-snapshot --volume-id volume_id --description "This is my volume snapshot."
```

volume_id is from step 2. **What is your command and its output?**

```
~ $ aws ec2 create-snapshot --volume-id vol-049e00e0e2d5824ce --description "This is my volume snapshot."
{
  ...skipping...
  {
    "Tags": [],
    "SnapshotId": "snap-04ff88f3eca9ee098",
    ...skipping...
  }
  {
    "Tags": [],
    "SnapshotId": "snap-04ff88f3eca9ee098",
    "VolumeId": "vol-049e00e0e2d5824ce",
    ...skipping...
  }
  {
    "Tags": [],
    "SnapshotId": "snap-04ff88f3eca9ee098",
    "VolumeId": "vol-049e00e0e2d5824ce",
    "State": "pending",
    "StartTime": "2025-01-31T22:15:49.588000+00:00",
    ...skipping...
  }
}
```

8. Check on the status of your snapshot via the following command:

```
aws ec2 describe-snapshots --snapshot-id snapshot_id
```

You can get *snapshot_id* from the output of step 7. **What is the output?** Check on the status of the snapshot until you see the snapshot's state transitions to "completed". The "completed" means the volume has been backed up.

```
~ $ aws ec2 describe-snapshots --snapshot-id snap-04ff88f3eca9ee098
{
  "Snapshots": [
    {
      "StorageTier": "standard",
      "TransferType": "standard",
      "CompletionTime": "2025-01-31T22:15:50.132000+00:00",
      "SnapshotId": "snap-04ff88f3eca9ee098",
      "VolumeId": "vol-049e00e0e2d5824ce",
      "State": "completed",
      "StartTime": "2025-01-31T22:15:49.588000+00:00",
      "Progress": "100%",
      "OwnerId": "881490118823",
      "Description": "This is my volume snapshot.",
      "VolumeSize": 1,
      "Encrypted": false
    }
  ]
}
```

9. Now let's create a new volume with the snapshot via the following:

```
aws ec2 create-volume --region us-east-1 --availability-zone us-east-1a --snapshot-id snapshot_id
```

What is your command and its output? You can get the *snapshot_id* from the output of step 7.

```

}
~ $ aws ec2 create-volume --region us-east-1 --availability-zone us-east-1a --snapshot-id snap-04ff88f3eca9ee098
{
  "Iops": 100,
  "Tags": [],
  "VolumeType": "gp2",
  "MultiAttachEnabled": false,
  "VolumeId": "vol-00f108ccb67a60528",
  "Size": 1,
  "SnapshotId": "snap-04ff88f3eca9ee098",
  "AvailabilityZone": "us-east-1a",
  "State": "creating",
  "CreateTime": "2025-01-31T22:18:51+00:00",
  "Encrypted": false
}

```

10. Check on the status of the volume. **What command did you run to check on the status? What is the output?**

```

~ $ aws ec2 describe-volume-status --volume-id vol-00f108ccb67a60528
{
  "VolumeStatuses": [
    {
      "Actions": [],
      "AvailabilityZone": "us-east-1a",
      "Events": [],
      "VolumeId": "vol-00f108ccb67a60528",
      "VolumeStatus": {
        "Details": [
          {
            "Name": "io-enabled",
            "Status": "passed"
          },
          {
            "Name": "io-performance",
            "Status": "not-applicable"
          }
        ],
        "Status": "ok"
      }
    }
  ]
}
~ $

```

```

~ $ aws ec2 describe-volumes --volume-id vol-00f108ccb67a60528
{
  "Volumes": [
    {
      "Iops": 100,
      "VolumeType": "gp2",
      "MultiAttachEnabled": false,
      "Operator": {
        "Managed": false
      },
      "VolumeId": "vol-00f108ccb67a60528",
      "Size": 1,
      "SnapshotId": "snap-04ff88f3eca9ee098",
      "AvailabilityZone": "us-east-1a",
      "State": "available",
      "CreateTime": "2025-01-31T22:18:51.089000+00:00",
      "Attachments": [],
      "Encrypted": false
    }
  ]
}

```

11. Attach the new volume to the instance via the following:

`aws ec2 attach-volume --volume-id volume_id --instance-id instance_id --device /dev/sdg`

The *volume_id* is from the output of step 9. The *instance_id* is from the output of step 1. **What is your command and its output?**

```

~ $ aws ec2 describe-volume-status --volume-id vol-00f108ccb67a60528
{
  "VolumeStatuses": [
    {
      "Actions": [],
      "AvailabilityZone": "us-east-1a",
      "Events": [],
      "VolumeId": "vol-00f108ccb67a60528",
      "VolumeStatus": {
        "Details": [
          {
            "Name": "io-enabled",
            "Status": "passed"
          },
          {
            "Name": "io-performance",
            "Status": "not-applicable"
          }
        ],
        "Status": "ok"
      }
    }
  ]
}
~ $ aws ec2 attach-volume --volume-id vol-00f108ccb67a60528 --instance-id i-02ecb5473d917941 --device /dev/sdg
{
  "VolumeId": "vol-00f108ccb67a60528",
  "InstanceId": "i-02ecb5473d917941",
  "Device": "/dev/sdg",
  "State": "attaching",
  "AttachTime": "2025-01-31T22:28:54.177000+00:00"
}

```

12. Switch to **the EC2 instance terminal**.
13. On the EC2 instance terminal, create a mount point /data2 and then mount the new volume to /data2. **What commands did you run? Please provide a screen capture of the commands you run.**

```

root@ip-172-31-42-75:~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
loop0        7:0    0  33.3M  1 loop /snap/amazon-ssm-agent/3552
loop1        7:1    0  55.5M  1 loop /snap/core18/1997
loop2        7:2    0  70.4M  1 loop /snap/lxd/19647
loop3        7:3    0  32.3M  1 loop /snap/snapd/11588
nvme0n1      259:0    0    8G  0 disk
└─nvme0n1p1  259:1    0    8G  0 part /
nvme1n1      259:2    0    1G  0 disk /data2
root@ip-172-31-42-75:~# cd /data2
root@ip-172-31-42-75:/data2# ls -l
total 20
drwx----- 2 root root 16384 Jan 31 22:33 lost+found
-rw-r--r-- 1 root root    24 Jan 31 22:43 mcy611.txt

```

14. Go to /data2. **Did you see mcy611.txt? Please provide a screen capture.**

```

root@ip-172-31-42-75:/data2# /mcy611.txt
bash: /mcy611.txt: No such file or directory
Hello, This is MCY 611!
~
~
~

```

15. Umount, detach, and destroy the volumes. The following is how I did:

```
root@ip-10-45-185-154:/data2# cd /
```

```
root@ip-10-45-185-154:/# umount /dev/nvme1n1
```

```
root@ip-10-45-185-154:/# umount /dev/nvme2n1
```

```
root@ip-172-31-42-75:/data2# cd
root@ip-172-31-42-75:~# umount /dev/nvme0n1
umount: /dev/nvme0n1: not mounted.
root@ip-172-31-42-75:~# umount /dev/nvme1n1
root@ip-172-31-42-75:~#
```

16. Switch to the **CloudShell terminal**. Detach and delete the first volume, which was created in step 2.

```
aws ec2 detach-volume --volume-id volume_id
```

Wait for 10 seconds so that the volume is detached. Then delete the volume via the following.

```
aws ec2 delete-volume --volume-id volume_id
```

The *volume_id* is from step 2.

Please detach and delete the second volume, which was created in step 9. **What commands did you run? What are the outputs?**

✕

Detach vol-049e00e0e2d5824ce?

After you detach a volume, you might still be charged for volume storage. If you no longer need the volume, delete it to stop incurring charges.

Are you sure that you want to detach volume vol-049e00e0e2d5824ce?

Cancel

Detach

✕

Delete vol-049e00e0e2d5824ce?

Volumes

⚠ After you delete a volume, its data is permanently deleted and the volume can no longer be attached to an instance.

Are you sure that you want to delete **vol-049e00e0e2d5824ce**?

To confirm deletion, type **delete** in the field.

Cancel

Delete

Delete the snapshot via the following:

```
aws ec2 delete-snapshot --snapshot-id snapshot_id
```

snapshot_id is from step 7.

Delete snap-0ec8c5ae2246eb359?

×

Are you sure that you want to delete snap-0ec8c5ae2246eb359?

Snapshots

To confirm deletion, type *delete* in the field.

delete

CancelDelete

17. Please terminate your EC2 instance.

What command did you run?

Terminate (delete) instance?

×

Instances

⚠ On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID

Termination protection

❏ i-02eccb5473d917941

✔ Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

CancelTerminate (delete)

Submission instructions

After completing this work, submit this word document with your answers to Canvas. In your word document, your answers should be in **bold** print. At the end of your word document, you should write a few sentences that explain the goals of this lab and a paragraph or two that summarizes (high level) the steps one needs to take in order to complete those goals.